

Set 1: Monthly Sales Data

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Monthly Sales Data
```

```
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Line Chart
```

```
plot(sales,
```

```
  type = "o",
```

```
  xaxt = "n",
```

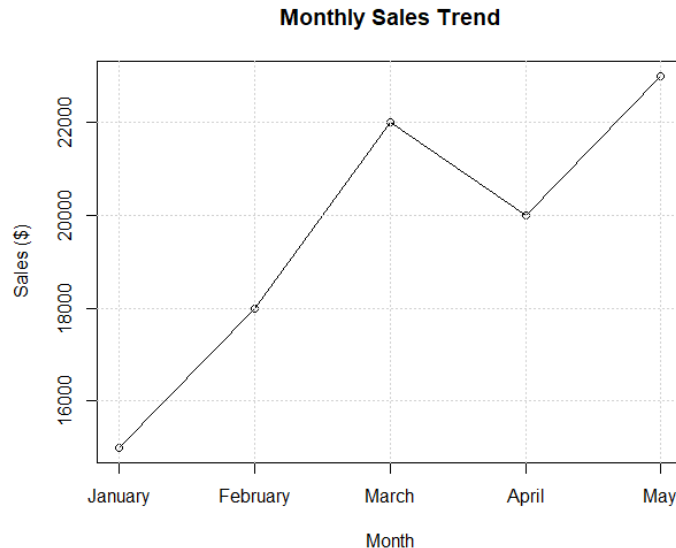
```
  xlab = "Month",
```

```
  ylab = "Sales ($)",
```

```
  main = "Monthly Sales Trend")
```

```
axis(1, at = 1:5, labels = month)
```

```
grid()
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

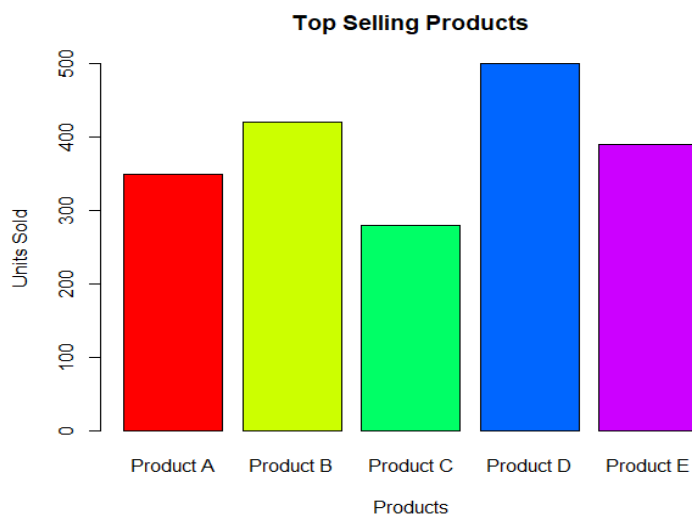
```
# Product Sales Data
```

```
products <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
product_sales <- c(350, 420, 280, 500, 390)
```

```
# Bar Chart
```

```
barplot(product_sales,  
        names.arg = products,  
        xlab = "Products",  
        ylab = "Units Sold",  
        main = "Top Selling Products",  
        col = rainbow(5))
```



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

```
# Data
```

```
advertising <- c(2000, 2500, 3000, 2800, 3500)
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

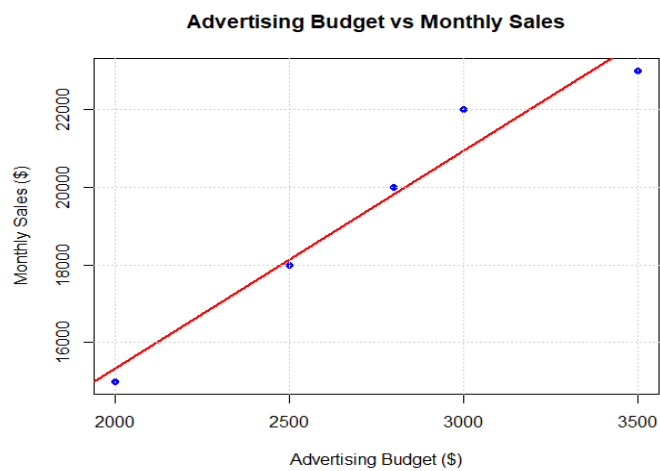
```
# Scatter Plot
```

```
plot(advertising,  
     sales,  
     pch = 19,  
     col = "blue",  
     xlab = "Advertising Budget ($)",  
     ylab = "Monthly Sales ($)",  
     main = "Advertising Budget vs Monthly Sales")
```

```
# Regression Line
```

```
abline(lm(sales ~ advertising), col = "red", lwd = 2)
```

grid()



Set 2: Customer Feedback Analysis

- 1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.**

```
# Customer Data
```

```
age <- c(25, 30, 35, 28, 40)
```

```
# Histogram
```

```
hist(age,
```

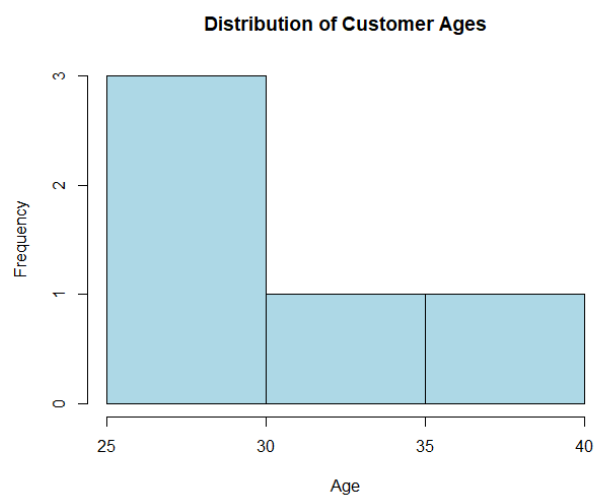
```
  main = "Distribution of Customer Ages",
```

```
  xlab = "Age",
```

```
  ylab = "Frequency",
```

```
  col = "lightblue",
```

```
  border = "black")
```



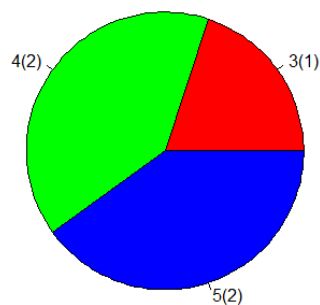
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
# Satisfaction Scores
satisfaction <- c(4, 5, 3, 4, 5)

# Count each score
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count),
                    "(", score_count, ")", sep = "" ),
    main = "Distribution of Customer Satisfaction Scores",
    col = rainbow(length(score_count)))
```

Distribution of Customer Satisfaction Scores



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# Customer Data
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create Age Groups
age_group <- cut(age,
    breaks = c(20, 30, 40, 50),
    labels = c("21-30", "31-40", "41-50"),
    include.lowest = TRUE)
```

```
# Frequency Table
```

```
data_table <- table(age_group, satisfaction)
```

```
# Stacked Bar Chart
```

```
barplot(data_table,
```

```
  beside = FALSE,
```

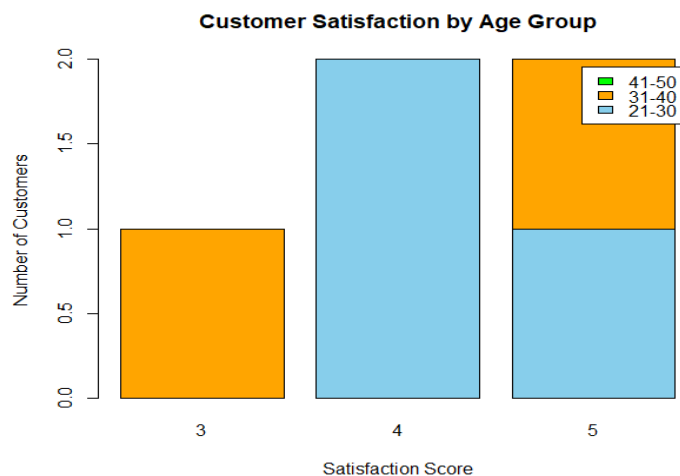
```
  col = c("skyblue", "orange", "green"),
```

```
  legend = rownames(data_table),
```

```
  main = "Customer Satisfaction by Age Group",
```

```
  xlab = "Satisfaction Score",
```

```
  ylab = "Number of  
Customers")
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Set 3: Employee Performance Evaluation

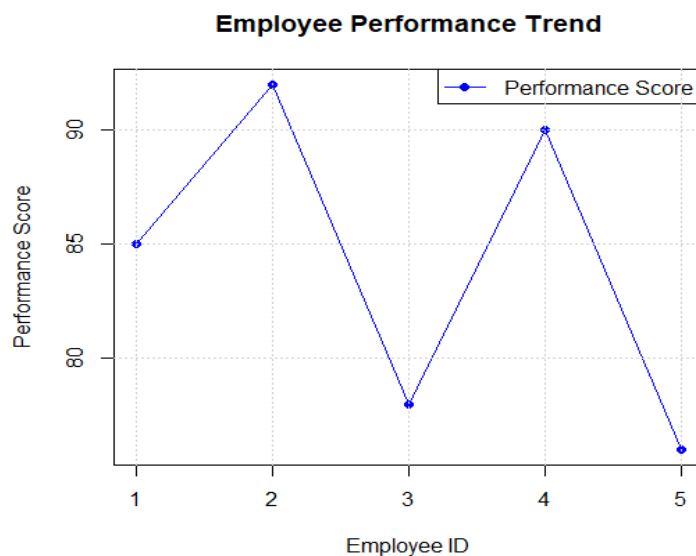
1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

```
# Employee Data
```

```
employee <- c(1, 2, 3, 4, 5)
```

```
performance <- c(85, 92, 78, 90, 76)
```

```
# Line Chart
plot(employee,
      performance,
      type = "o",
      pch = 16,
      col = "blue",
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
legend("topright",
      legend = "Performance Score",
      col = "blue",
      lty = 1,
      pch = 16)
grid()
```



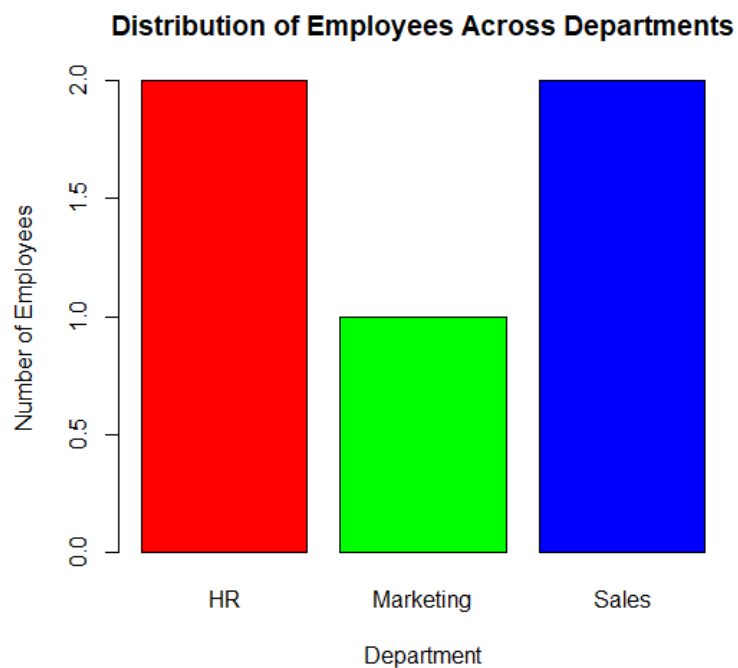
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Department Data
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

# Count Employees
dept_count <- table(department)
```

```
# Bar Chart
```

```
barplot(dept_count,  
        col = rainbow(length(dept_count)),  
        main = "Distribution of Employees Across Departments",  
        xlab = "Department",  
        ylab = "Number of Employees")
```



3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
# Data
```

```
years_service <- c(5, 3, 7, 4, 2)
```

```
performance <- c(85, 92, 78, 90, 76)
```

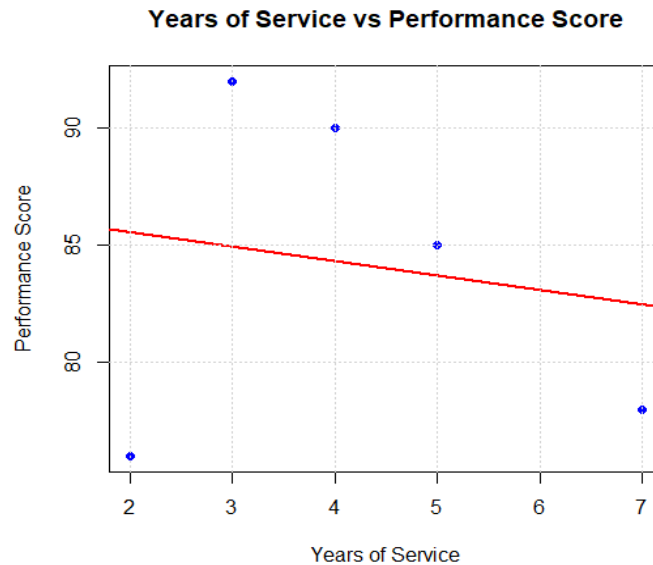
```
# Scatter Plot
```

```
plot(years_service,  
      performance,  
      pch = 19,  
      col = "blue",  
      xlab = "Years of Service",  
      ylab = "Performance Score",
```

```

    main = "Years of Service vs Performance Score")
# Regression Line
abline(lm(performance ~ years_service),
      col = "red",
      lwd = 2)
grid()

```



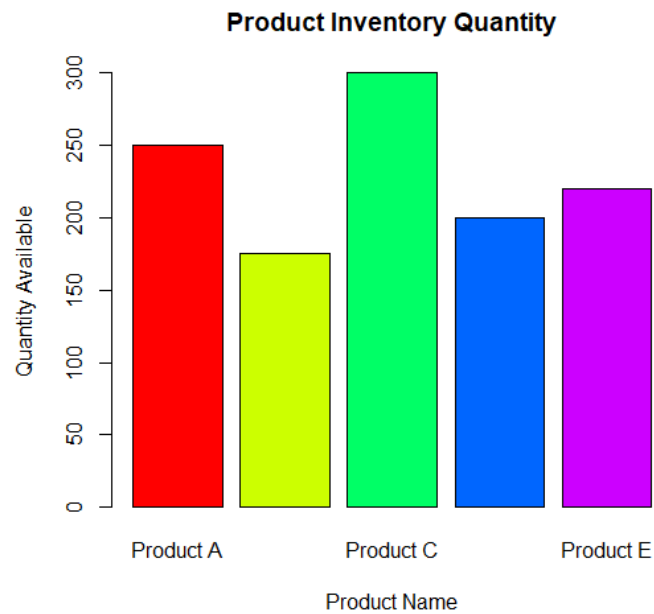
Set 4: Product Inventory Management

- 1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.**

```

# Product Data
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
# Bar Chart
barplot(quantity,
      names.arg = product,
      col = rainbow(5),
      xlab = "Product Name",
      ylab = "Quantity Available",
      main = "Product Inventory Quantity")

```

2. Generate a stacked bar chart to show the quantity of each product within different product categories.

Product Categories

```
category <- c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories")
```

Data Table

```
inventory <- data.frame(category, quantity)
```

Create Table

```
inventory_table <- xtabs(quantity ~ category + product)
```

Stacked Bar Chart

```
barplot(inventory_table,
```

```
  beside = FALSE,
```

```
  col = c("red", "blue"),
```

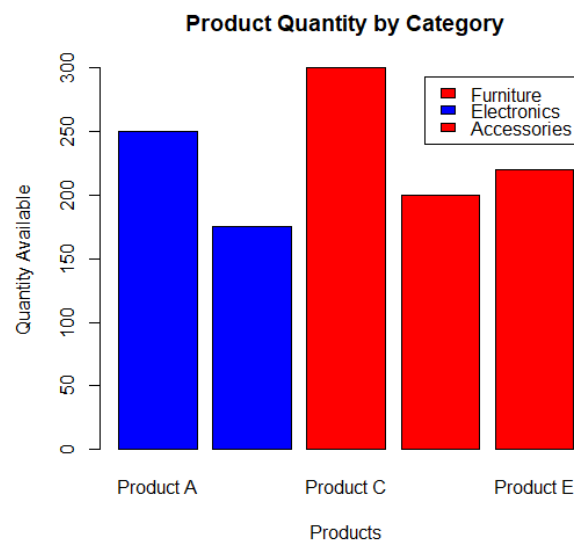
```
  legend = rownames(inventory_table),
```

```
  xlab = "Products",
```

```
  ylab = "Quantity Available",
```

```
  main =
```

```
  Quantity by
```



"Product
Category")

3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Data
```

```
price <- c(500, 750, 300, 650, 450)
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
# Scatter Plot
```

```
plot(price,
```

```
  quantity,
```

```
  pch = 19,
```

```
  col = "blue",
```

```
  xlab = "Product Price ($)",
```

```
  ylab = "Quantity Available",
```

```
  main = "Product Price vs Quantity Available")
```

```
# Regression Line
```

```
abline(lm(quantity ~ price),
```

```
  col = "red",
```

```
  lwd = 2)
```

```
grid()
```



Set 5: Website Analytics

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Website Traffic Data
```

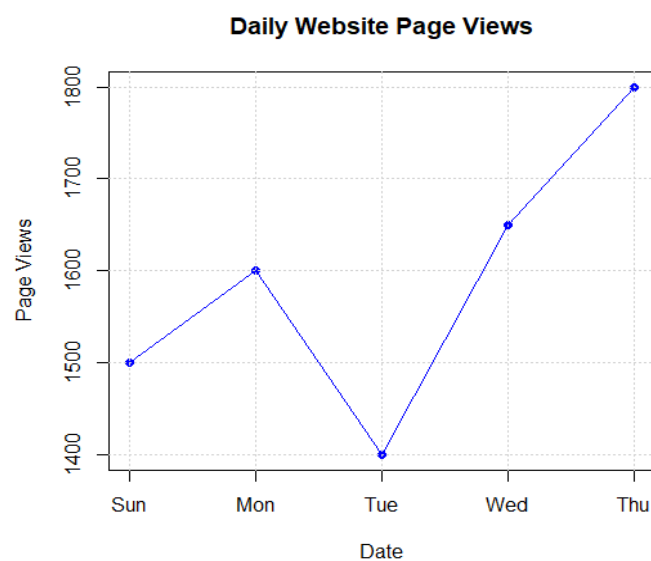
```
date <- as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03",  
                  "2023-01-04",  
                  "2023-01-05"))
```

```
page_views <- c(1500, 1600, 1400, 1650, 1800)
```

```
# Line Chart
```

```
plot(date,  
      page_views,  
      type = "o",  
      pch = 16,  
      col = "blue",  
      xlab = "Date",  
      ylab = "Page Views",  
      main = "Daily Website Page Views")
```

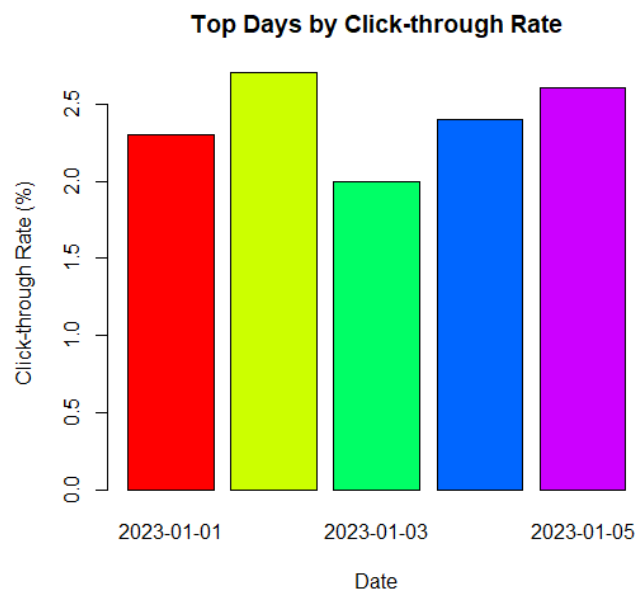
```
grid()
```



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Data
date <- c("2023-01-01",
          "2023-01-02",
          "2023-01-03",
          "2023-01-04",
          "2023-01-05")
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Bar Chart
barplot(ctr,
        names.arg = date,
        col = rainbow(5),
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top Days by Click-through Rate")
```



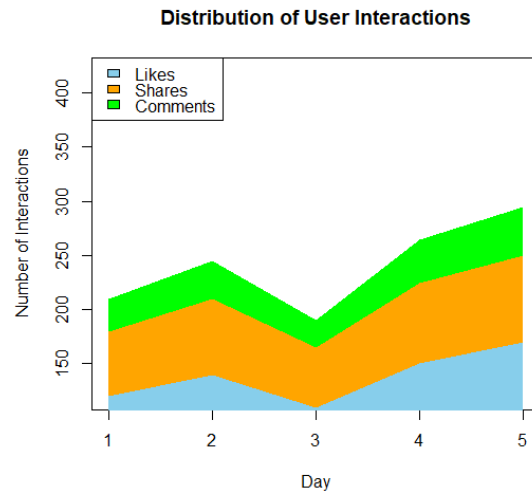
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```
# User Interaction Data
```

```

likes <- c(120, 140, 110, 150, 170)
shares <- c(60, 70, 55, 75, 80)
comments <- c(30, 35, 25, 40, 45)
# Matrix
interaction <- rbind(likes, shares, comments)
# Stacked Area Chart
x <- 1:5
plot(x,
     cumsum(interaction)[1:5],
     type = "n",
     xlab = "Day",
     ylab = "Number of Interactions",
     main = "Distribution of User Interactions")
polygon(c(x, rev(x)),
        c(rep(0,5), rev(likes)),
        col = "skyblue",
        border = NA)
polygon(c(x, rev(x)),
        c(likes, rev(likes + shares)),
        col = "orange",
        border = NA)
polygon(c(x, rev(x)),
        c(likes + shares, rev(likes + shares + comments)),
        col = "green",
        border = NA)
legend("topleft",
      legend = c("Likes", "Shares", "Comments"),
      fill = c("skyblue", "orange", "green"))

```



Set 6 : Product Sales Analysis

- Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.**

Monthly Product Sales Data

```
products <- c("Product A", "Product B", "Product C")
```

```
sales <- matrix(c(2000, 1500, 1200,
```

```
                2200, 1800, 1400,
```

```
                2400, 1600, 1100),
```

```
nrow = 3,
```

```
byrow = FALSE)
```

```
colnames(sales) <- products
```

```
rownames(sales) <- c("January", "February", "March")
```

Grouped Bar Chart

```
barplot(sales,
```

```
       beside = TRUE,
```

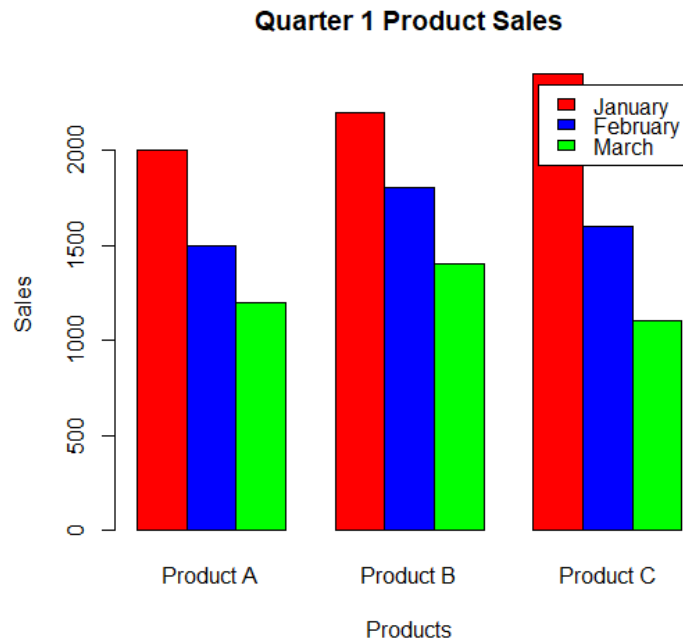
```
       col = c("red", "blue", "green"),
```

```
       main = "Quarter 1 Product Sales",
```

```
       xlab = "Products",
```

```
       ylab = "Sales",
```

```
       legend = rownames(sales))
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Sales Data

```
months <- c(1, 2, 3)
```

```
productA <- c(2000, 2200, 2400)
```

```
productB <- c(1500, 1800, 1600)
```

```
productC <- c(1200, 1400, 1100)
```

Cumulative Values

```
total1 <- productA
```

```
total2 <- productA + productB
```

```
total3 <- productA + productB + productC
```

Plot

```
plot(months, total3,
```

```
  type = "n",
```

```
  xlab = "Month",
```

```
  ylab = "Sales",
```

```
  main = "Overall Sales Trend")
```

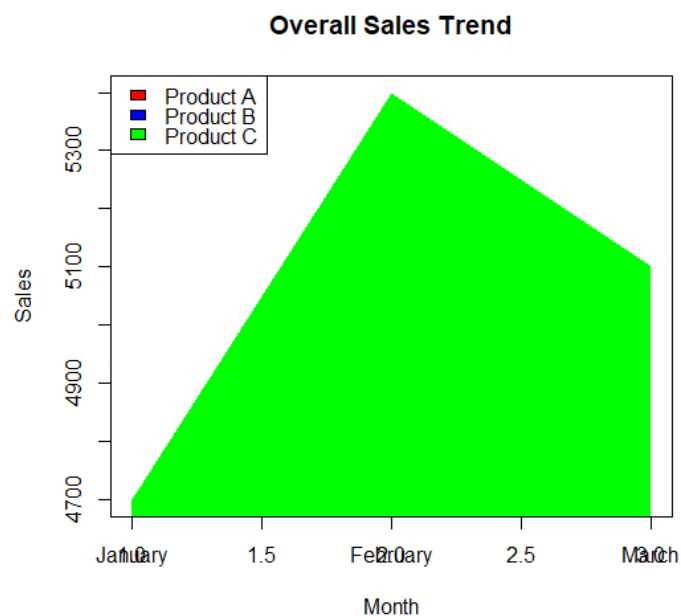
```
polygon(c(months, rev(months)),
```

```
  c(rep(0,3), rev(total1)),
```

```

col = "red",
border = NA)
polygon(c(months, rev(months)),
c(total1, rev(total2)),
col = "blue",
border = NA)
polygon(c(months, rev(months)),
c(total2, rev(total3)),
col = "green",
border = NA)
axis(1,
at = months,
labels = c("January", "February", "March"))
legend("topleft",
legend = c("Product A", "Product B", "Product C"),
fill = c("red", "blue", "green"))

```



3. Build a table to show the monthly sales data for each product. Label the table elements.

Create Data Frame


```

sales_table <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
# Display Table
print(sales_table)

```

	Product	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Set 7 Customer Demographics Analysis

- 1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.**

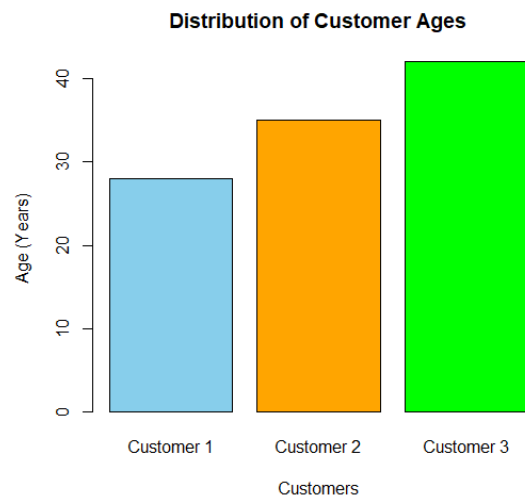
```

# Customer Data
age <- c(28, 35, 42)

# Customer IDs
customer <- c("Customer 1", "Customer 2", "Customer 3")

# Bar Chart
barplot(age,
  names.arg = customer,
  col = c("skyblue", "orange", "green"),
  xlab = "Customers",
  ylab = "Age (Years)",
  main = "Distribution of Customer Ages")

```



2. Generate a pie chart to represent the distribution of customers by gender.

Gender Data

```
gender <- c("Female", "Male", "Female")
```

Count Gender

```
gender_count <- table(gender)
```

Pie Chart

```
pie(gender_count,
```

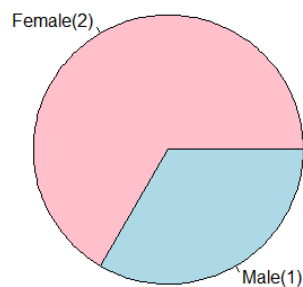
```
  labels = paste(names(gender_count),
```

```
    "(", gender_count, ")", sep = "" ),
```

```
  col = c("pink", "lightblue"),
```

```
  main = "Customer Distribution by Gender")
```

Customer Distribution by Gender



3. Build a table to show the demographic information for each customer. Label the table elements.

Customer Demographic Table

```
customer_data <- data.frame(
```

```

Customer_ID = c(1, 2, 3),
Age = c(28, 35, 42),
Gender = c("Female", "Male", "Female"),
Income = c(50000, 60000, 75000)
)

# Display Table
print(customer_data)

```

Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Set8 Employee Performance Analysis

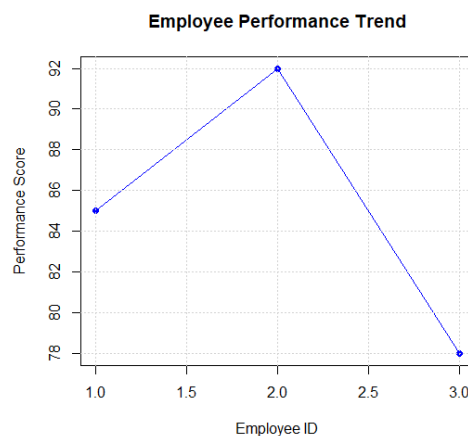
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```

# Employee Data
employee <- c(1, 2, 3)
performance <- c(85, 92, 78)

# Line Chart
plot(employee,
      performance,
      type = "o",
      pch = 16,
      col = "blue",
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
grid()

```

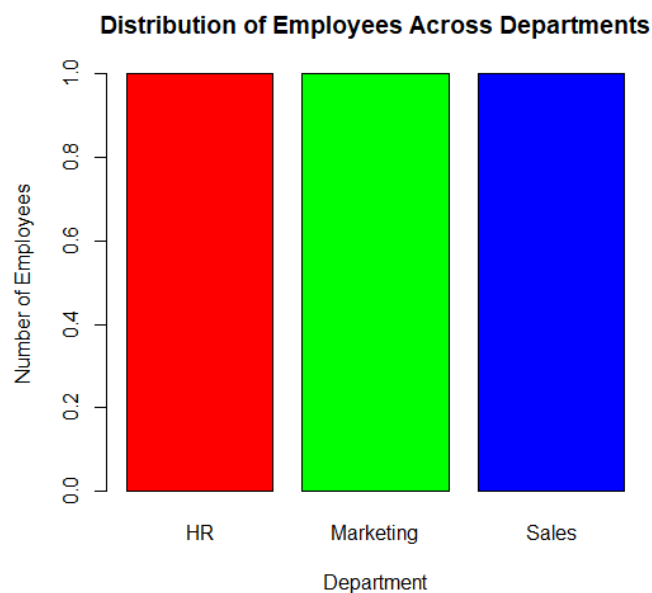


2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Department Data
department <- c("Sales", "HR", "Marketing")

# Count Employees
dept_count <- table(department)

# Bar Chart
barplot(dept_count,
        names.arg = names(dept_count),
        col = c("red", "green", "blue"),
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Distribution of Employees Across Departments")
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
# Employee Performance Table
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
```

```

Years_of_Service = c(5, 3, 7),
Performance_Score = c(85, 92, 78)
)
# Display Table
print(employee_data)

```

Employee_ID	Department	Years_of_Service	Performance_Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

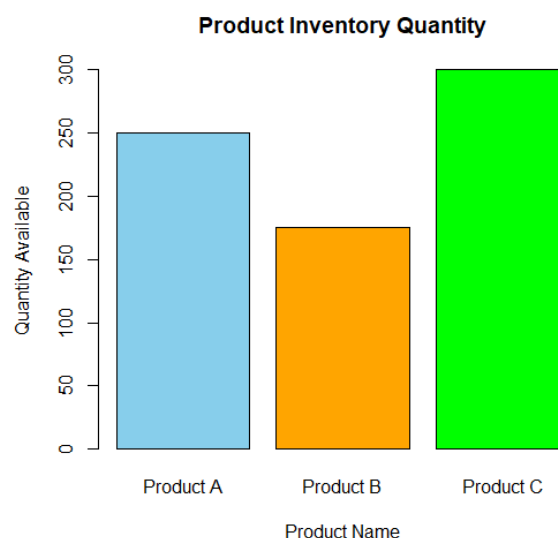
Set 9 Product Inventory Management

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```

# Product Data
product <- c("Product A", "Product B", "Product C")
quantity <- c(250, 175, 300)
# Bar Chart
barplot(quantity,
names.arg = product,
col = c("skyblue", "orange", "green"),
xlab = "Product Name",
ylab = "Quantity Available",
main = "Product Inventory Quantity")

```

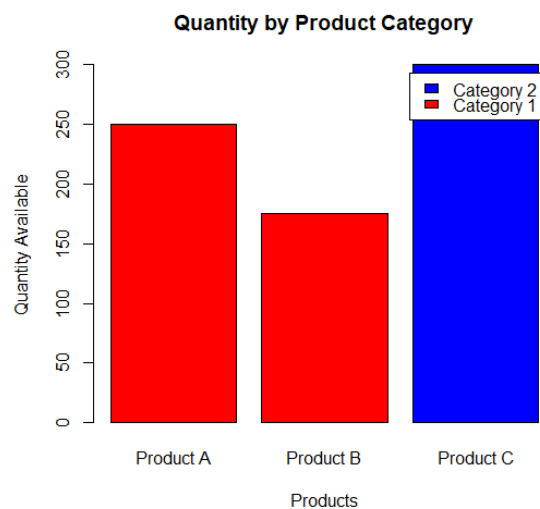


2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Product Data
product <- c("Product A", "Product B", "Product C")
category <- c("Category 1", "Category 1", "Category 2")
quantity <- c(250, 175, 300)

# Create Table
inventory_table <- xtabs(quantity ~ category + product)

# Stacked Bar Chart
barplot(inventory_table,
        beside = FALSE,
        col = c("red", "blue"),
        legend = rownames(inventory_table),
        xlab = "Products",
        ylab = "Quantity Available",
        main = "Quantity by Product Category")
```



3. Build a table to show the inventory data for each product. Label the table elements.

```
# Inventory Table
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
```

```

Price = c(20, 15, 18)
)

# Display Table
print(inventory_data)

```

Product_ID	Product_Name	Quantity_Available	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Set 10: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Label the chart elements.

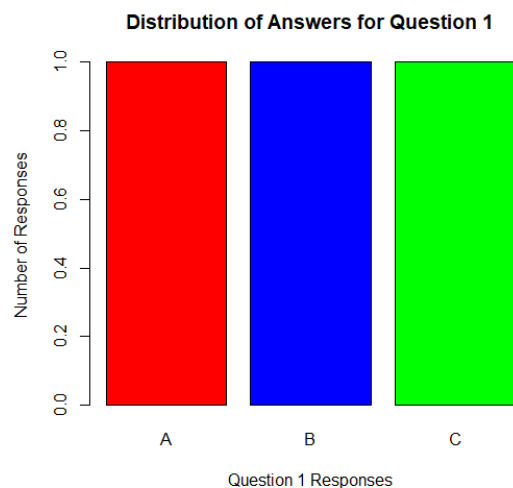
```

# Survey Data
responses <- c("A", "B", "C")

# Count Responses
response_count <- table(responses)

# Grouped Bar Chart
barplot(response_count,
  beside = TRUE,
  col = c("red", "blue", "green"),
  names.arg = names(response_count),
  xlab = "Question 1 Responses",
  ylab = "Number of Responses",
  main = "Distribution of Answers for Question 1")

```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

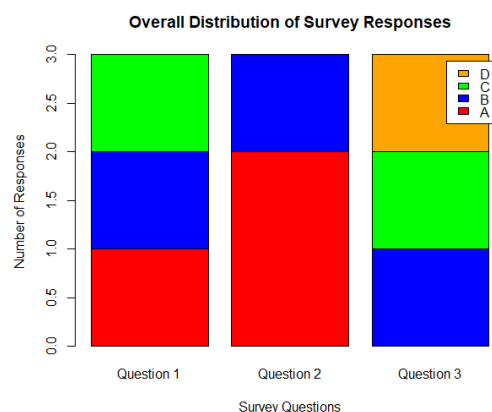
```
# Survey Data
survey_data <- data.frame(
  Q1 = c("A", "B", "C"),
  Q2 = c("B", "A", "A"),
  Q3 = c("C", "D", "B")
)

# Response Levels
levels <- c("A", "B", "C", "D")

# Frequency Table
response_matrix <- rbind(
  table(factor(survey_data$Q1, levels = levels)),
  table(factor(survey_data$Q2, levels = levels)),
  table(factor(survey_data$Q3, levels = levels))
)

rownames(response_matrix) <- c("Question 1", "Question 2", "Question 3")

# Stacked Bar Chart
barplot(t(response_matrix),
  beside = FALSE,
  col = c("red", "blue", "green", "orange"),
  xlab = "Survey Questions",
  ylab = "Number of Responses",
  main = "Overall Distribution of Survey Responses",
  legend.text = c("A", "B", "C", "D"))
```



3. Build a table to show the survey response data for each survey. Label the table elements.

```
# Survey Response Table  
survey_table <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)
```

```
# Display Table
```

```
print(survey_table)
```

Survey_ID	Question_1	Question_2	Question_3	
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B